Telecom Fraud Radar — Project Status and Next Steps

Prepared 16 October 2026 from the repository, Steps.odt, deepseek.txt, and the latest LocalStack work. This status note supplements Steps.odt; it does not replace it.

Project goal and constraints

Build an AWS-first, multi-source telecom fraud detection portfolio project that runs locally on LocalStack. No Kubernetes, paid AWS cloud, payment card, or real subscriber data. All records and labels are simulated. Keep the README explicit about LocalStack and synthetic data.

The first demo uses SIM activation (source 1), revenue (2), voice/data usage (3), bundle purchases (5), and subscriber profile (8). Bundle usage (6) and churn/customer state (7) remain deferred. Bundle catalog (4) is reference data.

Progress at a glance

Week 1 — Foundations: COMPLETE. Docker Compose / LocalStack setup, Makefile, S3 bronze bucket and raw prefixes, DynamoDB reference tables, Kinesis streams, SNS topic, Pydantic contracts, requirements, and the initial honest README are present.

Week 2 — Simulator and bronze: IMPLEMENTED, with a reproducibility gap to fix. The repository has baseline generators, four fraud injectors, overlap/hard cases, separate labels, S3/Kinesis/DynamoDB writers, and activation/purchase Lambda handlers. Raw and Silver data are present in the current LocalStack state. However, simulator/run.py currently does not invoke writers.write_all(), so make simulate alone does not reproduce the AWS writes claimed by its target.

Week 3 — Quality gate and Silver: COMPLETE in the captured chat and current code. Schema, duplicate, range, referential checks, row quarantine, per-source metrics, freshness alerts, and a Glue-compatible wrapper exist. Raw inputs are processed into silver/ and quarantine/; metrics go to quality_metrics/ and threshold/staleness alerts use SNS.

Week 4 — Gold, Athena, investigation rules: COMPLETE. `make week4` ran end to end successfully. Gold wrote 2,500 account-day rows and exported 500 subscriber profiles plus 8 bundles. Athena created all eight external tables and registered partitions. All four investigation queries ran successfully. The rules workflow upserted 26 candidate alerts.

Weeks 5–6 — Model/scoring and dashboard/demo: NOT STARTED. No model training pipeline, model artifact bucket, scoring Lambda, dashboard export/UI, demo video, or LinkedIn draft is present yet.

Week 4 work completed in this session

Gold transform: transform/gold.py reads the five Silver datasets, deduplicates stable source IDs across repeated quality-gate runs, aggregates one row per subscriber and event date, and writes Parquet under gold/account_daily_features/year=YYYY/month=MM/day=DD/. It also exports DynamoDB subscriber and bundle snapshots to gold/reference/ for Athena joins.

Observed LocalStack result using the small dataset: 2,500 account-day rows written across five date partitions; 500 subscriber profiles and 8 bundle catalog entries exported. Bucket: telco-fraud-bronze.

Athena assets: transform/athena/tables.sql defines eight Parquet tables over Silver, Gold, and exported reference snapshots. transform/athena/setup.py provisions Glue database telecom_fraud_radar, Athena workgroup fraud-radar-workgroup, DynamoDB fraud_alerts (alert_id key, subscriber-index GSI), table definitions, and discovered S3 partitions. Four saved investigation SQL files cover the fraud patterns.

Rules: scoring/rules.py evaluates four explainable rules and upserts idempotently keyed candidates in fraud_alerts. Final `make week4` wrote 26 candidates: 4 SIM_SWAP_INTL_24H, 14 IRSF_HIGH_RISK_BURST, 5 NEW_ACCOUNT_HIGH_VALUE_PURCHASE, and 3 REPEATED_BUNDLE_PURCHASES.

Athena debugging found two schema issues and both are fixed: the physical Parquet column remains named `timestamp`, so Athena DDL quotes it and investigation SQL aliases it as `event_timestamp`; DynamoDB reference exports store whole-number fields as decimal(38,0), now reflected in the two reference table definitions. The setup refresh drops and recreates external table metadata only; S3 Parquet objects are preserved. Final `make week4` completed successfully.

Files added or updated

Added: transform/gold.py, transform/athena/tables.sql, transform/athena/setup.py, transform/athena/query.py, four files under transform/athena/investigations/, scoring/rules.py, scoring/rules_config.yaml, and package initializers.

Updated: Makefile has gold, athena, athena-query, rules, and week4 targets. README now marks Week 4 complete and records the verified LocalStack run. simulator/config.yaml already had user modifications before this work and was not edited.

What remains to finish Week 4

1. Week 4 validation: all four saved queries succeeded. They returned 4 SIM-swap matches, 6 dated IRSF groups, 4 high-value purchase candidates, and 3 repeat-purchase groups.

2. `make week4` completed: it rebuilt Gold, recreated the Athena catalog and partitions, and wrote 26 rule candidates to DynamoDB.

3. Week 5 is next: build leakage-safe features, evaluate a rules-only baseline, train and evaluate a small gradient-boosting model, then add S3 model storage and a scoring Lambda.

4. Week 6 follows: alerts/quality JSON export, static dashboard, short demo video, README polish, and LinkedIn post.

5. Preserve the stated limits: all records are simulated; repeat purchases are only a bundle-abuse proxy until source 6 exists; rule candidates are not automatic account blocks.

Known limitations and assumptions to keep honest

Bundle-usage source 6 is deferred. The implemented bundle rule detects repeated purchases (3+ of one bundle in a rolling 24-hour window), not consumption over allowance. Do not describe it as confirmed bundle over-consumption detection.

Subscription-fraud rule uses available profile/purchase/payment signals; it is a candidate heuristic, not a calibrated production policy. Rules do not auto-block accounts and do not publish SNS alerts yet; those are for the later scoring/alerting stage.

Gold is account-day grain. The exact SIM swap → international calls within 24 hours rule uses event timestamps from Silver, not daily aggregates.

Revenue Silver currently covers raw/revenue/prepaid only; postpaid simulation and ingestion are not represented in the present runner.

LocalStack Athena first-use startup downloaded large Java/Hadoop dependencies and took several minutes. LocalStack documentation currently describes Athena as plan-gated; if the running token does not include it, keep the SQL/catalog assets, report that the local plan cannot execute Athena, and do not switch to paid AWS. Official reference: https://docs.localstack.cloud/aws/services/athena/

Do not claim this is production-validated or trained on real telecom data. Kaggle validation, model training, model-serving cold-start measurements, PII hashing, least-privilege IAM, and retention policies are not implemented.

Next project stages

Week 5 — Feature engineering and model: build account-level features without joining labels into predictors; split/evaluate by subscriber to avoid leakage; report precision/recall and PR-AUC for the imbalanced simulated population; compare rules-only and model-assisted candidates. Train a small gradient-boosting model on simulator labels first. Kaggle is optional secondary validation only if its fields and label semantics map honestly. Save model artifact to S3 and implement a scoring Lambda that writes alerts to DynamoDB and publishes SNS.

Week 6 — Portfolio delivery: export alerts and quality metrics as JSON, build the static dashboard, write/record the 30-second demo, update README with tested LocalStack commands and limitations, and prepare a LinkedIn post.

Current status and next action

Week 4 is complete. Start Week 5 feature engineering and model evaluation.

To rerun the validated Week 4 workflow with LocalStack running, execute: make week4.

The setup recreates Athena external-table metadata and re-registers partitions; it does not delete S3 Parquet data.